MARITIME (MRT)

Market-First → Maritime Payment Rail (Escrow) Roadmap & Workflows

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Purpose

This roadmap starts with markets (traders, liquidity, transparency) and transitions—step by step—toward a maritime payment rail.

It defines practical workflows for Time Charter and Voyage Charter using an escrow model that keeps funds secure while claims are resolved.

It is designed to work alongside Charter Parties (CP). It does not attempt to replace CP clauses or legal interpretation.

Core message

The Charter Party remains the contract of record.

We add a safer payment path: funds are held in an escrow vault until performance is confirmed or a dispute is resolved.

This is a payment rail that secures funds—not a new legal system.

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1) Market-first: what exists today, and why it matters

The project starts with an openly traded ERC-20 asset and a transparent market surface: on-chain verification, public documentation, and DEX liquidity formation. The purpose of this phase is to build a credible baseline: discoverable price formation, reduced slippage over time, and a public record of how liquidity and wallets are managed.

Outputs of the market phase

Beginner materials (how to verify, how to buy, how to avoid counterfeit assets).

Transparency artifacts (official wallets, allocations, reward budgets, on-chain proof).

Liquidity incentives that reward liquidity + time (not trading volume) to avoid wash-trading incentives.

Why markets first?

- Without a functioning market, it is hard to attract feedback and early adopters for utility modules.
- A minimal token layer and public proof reduce the hidden-controls risk perception.
- Escrow design must be informed by real workflows; the market phase is where you learn what users actually need.

2) Payment rail explained: what escrow is, and where it lives

Escrow is a neutral vault. Funds do not sit with Owner or Charterer. Funds are locked in a smart-contract vault on Polygon and released according to agreed actions.

Payment rail framing

CP clauses remain off-chain and enforceable the usual way.

We change the payment path: stablecoin funds sit in a controlled escrow vault.

The goal is practical: reduce payment delays and reduce 'who holds the money' disputes.

Where is the escrow?

- On-chain: Polygon smart contract vault holding USDC.
- Off-chain: a simple dApp used to create deposits, propose settlements, and approve releases.
- Documents stay off-chain; hashes may be anchored on-chain when needed for audit trail.

3) Trust model: why neither side can run away with the money

Shipping disputes escalate when one side can unilaterally withhold or seize funds. Escrow prevents that. Funds are held in a neutral vault and can only move based on an agreed settlement or a defined resolution process.

Practical default: 2-of-3 approvals

Three signers: Owner, Charterer, and an independent third party (broker/agent/arbitrator). Any release requires two approvals.

Deposit USDC	Escrow vault locks funds	Settlement approval (2-of-3)	Payout / Refund

What this prevents

Charterer cannot 'deduct first and argue later' by simply keeping the cash.

Owner cannot force payment without acknowledgement of performance or a resolution.

In a dispute, funds remain locked until the settlement is agreed or resolved.

4) Roadmap: phases, deliverables, and risk controls

This roadmap moves from markets to escrow in controlled steps. The aim is adoption without over-promising.

Phase	Goal	Deliverables (summary)	Risk controls
Phase 1 — Markets (0–2 months)	Credibility and access	Beginner guides (EN/TR), verification checklist, wallet proofs, transparent LP incentives	Small-scale testing, on-chain proof links, consistent disclosures
Phase 2 — Escrow MVP (2–4 months)	Secure payment rail (pilot)	Minimal dApp: create vault, deposit USDC, propose settlement split, approve (2-of-3)	Pilot caps, USDC allowlist, multisig approvals, independent review
Phase 3 — Charter templates (4–6 months)	Time/Voyage workflows	Time: hire drip + off-hire freeze. Voyage: freight milestones + laytime vault	Template clarity, evidence checklists, conservative timeouts
Phase 4 — Claims & reporting (6–9 months)	Claims reserve + audit trail	Claims reserve vault, dispute logs, report-hash anchoring (carbon/proof artifacts)	Privacy-by-design, limited data on-chain, verifier roles
Phase 5 — Integrations (9–12 months)	Operational adoption	APIs, accounting exports, partner pilots, internal controls	Gradual rollout, SOPs, monitoring, incident response

Safe transition principle

Start with low-risk pilots (small caps) before touching full hire/freight flows.

Avoid automatic timeout release at the beginning; use explicit approvals and clear procedures.

Measure disputes per transaction, time-to-resolution, and payout accuracy. Scale only after stability is proven.

5) Time Charter workflows

Objective: secure hire payments while giving both parties a fair way to handle off-hire and performance adjustments without turning payment into leverage.

5.1 Base model: USD 15,000 per 7 days (Hire Escrow + daily drip)

Charterer deposits 15,000 USDC	Hire vault	Daily drip to Owner	Period settlement

Why drip is practical

Charterer pays in advance into a neutral vault (similar intent to hire paid in advance).

Owner receives hire gradually through the period, reducing payment friction.

If a dispute is raised, the remaining balance is frozen without reversing already-earned hire by default.

5.2 Off-hire event

- Charterer raises an off-hire claim → the vault pauses the drip and freezes the remaining balance.
- Parties exchange evidence (logbook, noon reports, class attendance, notices).
- Resolution: full hire payable (release to Owner) or accepted off-hire time (refund agreed portion to Charterer).

5.3 Performance (speed/consumption) claims

Operational approach: hold a small claims reserve per period (e.g., 5%). Settle periodically (month-end or voyage-end).

- Retain 5% of each hire period in a Claims Reserve vault.
- After agreement, pay out or refund the reserve accordingly.

5.4 Bunkers (delivery/redelivery) and bunker disputes

- Use a separate Bunker Adjustment vault for delivery/redelivery surveys and price references.
- If contamination/off-spec is alleged, freeze the bunker vault portion until lab results and settlement are agreed.

6) Voyage Charter workflows

Objective: secure freight via milestone releases, and keep laytime/demurrage settlement separate so one disagreement does not block the entire freight.

6.1 Freight milestones (example 20/60/20)

Charterer deposits USDC	Freight vault	BL issued → 60%	Discharge complete → 20%	Closeout

Why milestone releases work

Voyage performance happens in stages. Payments released by stage align incentives and reduce end-of-voyage friction.

If a dispute arises, you can freeze only the relevant remainder rather than block the whole payment.

6.2 Laytime / Demurrage / Despatch

Do not mix this with freight. Use a dedicated Laytime Settlement vault. Parties agree a figure off-chain (SOF-based) and settle via the vault.

- SOF and documents remain off-chain; hashes may be anchored on-chain for audit trail.
- If agreed → pay/refund immediately; if disputed → freeze and resolve via approvals (2-of-3) or arbitrator decision.

6.3 Laycan miss / cancellation

- If laycan is missed and the voyage is cancelled: unreleased freight remains in the vault and is refunded.
- If CP includes cancellation fee or deadfreight logic: apply it as a parameterized split after agreement/decision.

7) Claims & dispute catalog (event → vault behavior → evidence)

The vault does not decide who is right. It enforces neutral custody of funds and executes an agreed settlement (or a defined resolution).

Event	Where it appears	Vault behavior	Common evidence
Engine breakdown / loss of performance	Time & Voyage	Raise dispute → pause drip (Time) / pause remaining milestones (Voyage)	Noon reports, engine log, class attendance
Off-hire claim	Time	Pause drip + freeze remaining balance; settle by agreed time ratio or full payment	Notices, logs, statements
Speed/consumption claim	Time	Use claims reserve; settle periodically	Performance calculations, voyage data
Laycan miss / cancellation	Voyage	Do not release freight; refund or apply agreed fee split	CP clause, ETA/NOR records
Demurrage/Despatch dispute	Voyage	Freeze laytime vault until settled	SOF, NOR validity, port logs
Cargo damage/shortage	Time & Voyage	Use claims reserve; release remainder; settle claim later	Survey reports, BL remarks, P&I; letters
Port closure / strike / war risk	Time & Voyage	Template-based: continue payments or freeze remainder; extra costs via separate vault	Circulars, notices, port info
Bunker contamination /off-spec	Time & Voyage	Freeze bunker vault; settle on lab results	BDN, lab tests, survey notes

Design principle

The vault is a custody and execution layer. Legal interpretation stays with CP, evidence, and the agreed resolution process.

The value proposition is that funds cannot be used as leverage during negotiation.

8) Technical architecture (minimum viable stack)

Start simple. A minimal design is easier to review and harder to break.

8.1 On-chain modules (minimum)

- EscrowFactory: creates escrow vault instances with template parameters.
- EscrowVault: deposit, propose settlement split, approve (2-of-3), freeze/unfreeze, execute payout.
- USDC allowlist: accept only the intended USDC token address to reduce counterfeit risk.
- Events: every action emits an on-chain event for audit trail.

8.2 Off-chain modules (minimum)

- A simple dApp: Create, Deposit, Propose, Approve.
- Evidence storage: documents remain off-chain with optional hash anchoring.
- Reporting: weekly/monthly stats (settled, disputed, resolved; median resolution time).

Security baseline (MVP)

Small pilot caps and staged rollout.

2-of-3 approvals with a reputable third signer.

No aggressive automations at first (avoid timeouts).

Clear operational SOP: what evidence is required for common disputes.

9) Operational rollout (pilots, limits, metrics)

Adoption in shipping is conservative. The goal is not to replace banks overnight; it is to reduce friction and improve certainty where it hurts most.

9.1 Pilot strategy

- Start with a low-risk use case: reserve/deposit vault or laytime settlement vault.
- Use one real counterparty pair (Owner + Charterer) plus one independent signer (broker/agent).
- Track: payment timeliness, dispute frequency, time-to-resolution, and user satisfaction.

9.2 Scale-up path

- Phase A: reserve/deposit usage → Phase B: partial hire/freight → Phase C: full flows.
- Keep the process explicit at the beginning; automate only after patterns are proven.
- Add integrations (accounting exports, APIs) once operational confidence is established.

How to communicate this (non-hype)

We do not replace the Charter Party; we secure the payment path.

Stablecoin settlement reduces volatility and simplifies accounting.

On-chain audit trail improves transparency without exposing sensitive documents.

10) Appendices

10.1 Template — Time Charter (summary parameters)

Time Charter template (example)

Period: 7 days | Hire: 15,000 USDC | Mode: advance deposit + daily drip

Off-hire: claim → pause drip → freeze remaining balance

Performance: 5% claims reserve | Settlement: month-end

Bunkers: separate adjustment vault (delivery/redelivery)

Dispute approvals: 2-of-3 (Owner + Charterer + Agent)

10.2 Template — Voyage Charter (summary parameters)

Voyage template (example)

Freight milestones: 20% arrival/NOR / 60% BL issued / 20% discharge complete

Laytime: separate settlement vault (SOF-based)

Claims reserve: optional 10% for cargo-related exposure

Laycan miss/cancel: refund unreleased freight; apply agreed fee split if applicable

Dispute approvals: 2-of-3

10.3 Glossary (very short)

- Escrow vault: neutral custody contract holding funds until release/refund is approved.
- Milestone: a release stage (BL issued, discharge complete, etc.).
- Laytime/Demurrage: time accounting and delay payment settlement.
- Off-hire: time charter period where vessel is considered off-hire under CP clauses.
- 2-of-3: two approvals out of three signers required to execute payout/refund.